Wildfire Analysis & Prediction for California

Course: IS-517: Methods of Data Science
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Motivation

Wildfires in California are a major threat to life, property, and the environment.
This project aims to identify key environmental factors influencing wildfire duration, classify unknown causes, and detect susceptible zones using machine learning models.

Dataset Description

We used two primary datasets:

(1) **CALFIRE wildfire perimeters from 2008 to 2023**, and

(2) **an engineered dataset from Zenodo.org** with daily fire occurrence and weather data.

Data engineering steps included calculating fire duration, extracting latitude/longitude from shapes, and integrating temperature and wind data from APIs for modeling.

Initial Inferences from Dataset

Field Correlation

Correlation between features was generally low—Pearson correlation (Figure 1 below), indicating the need for nonlinear classification methods.

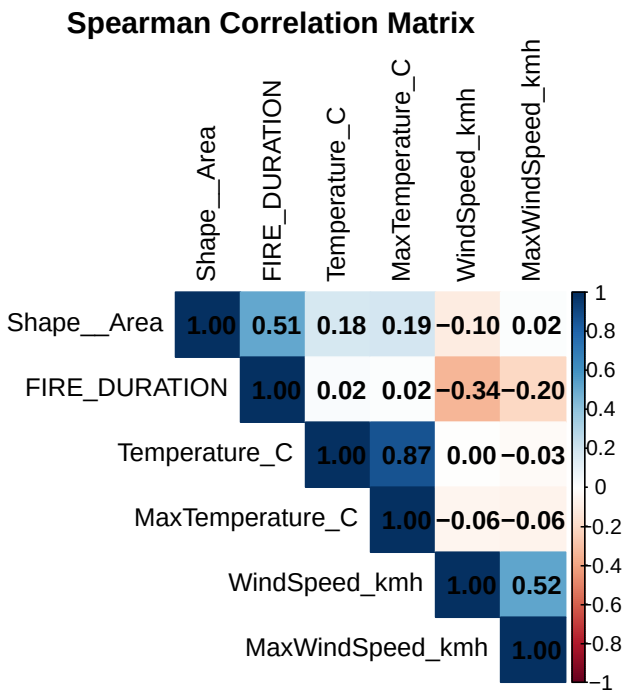

Figure 1: Spearman correlation matrix

Seasonal Trends

Strong seasonal trend observed in histogram plots. Figure 2 shows significant seasonal fire activity during summer due to high temperatures.

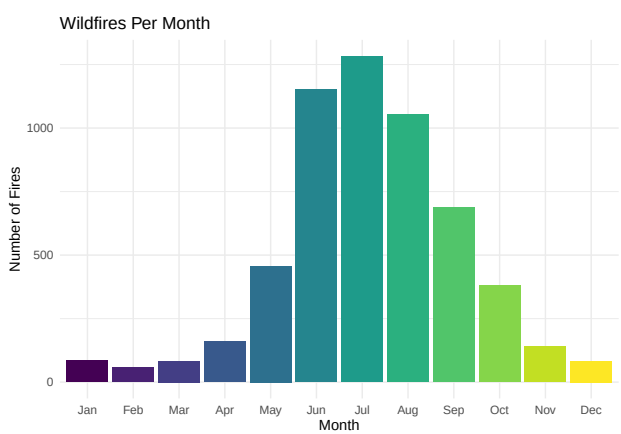

Figure 2: Seasonal trends

Fire Distribution Trends by Days

From the figure-3 below it can be seen that most of the fires are usually contained within the same day which leads to a very large class of fires being 0 days.

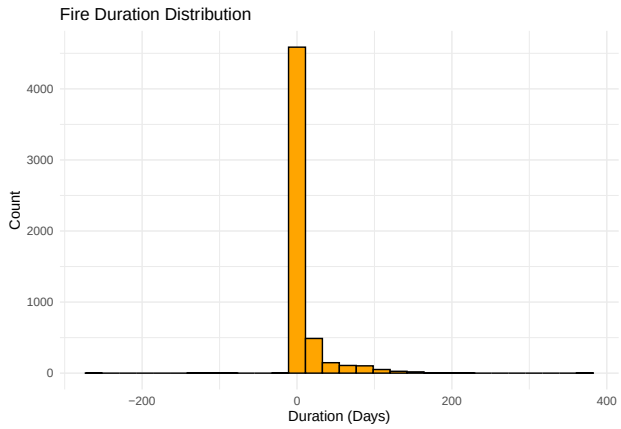

Figure 3: Fire duration distribution

Classification of Fires by Cause

Figure-4 below shows that most of the fires that occur are unknown causes of fire around 31%. This is huge chunk of data that needs to be studied to attempt and identify the causes of fire.

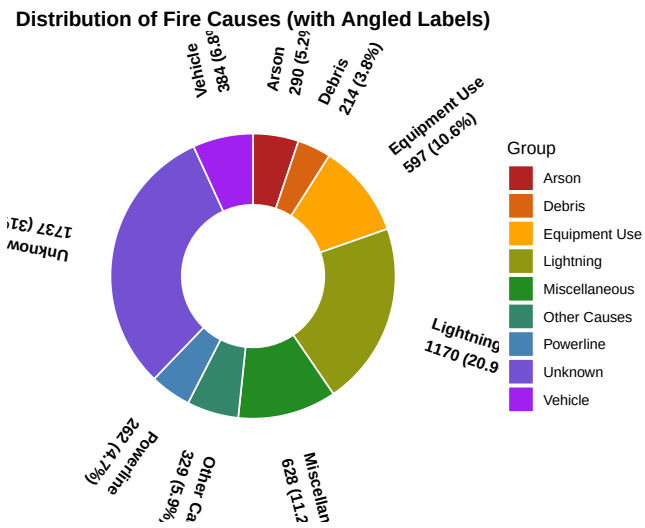

Figure 4: Classification of Fires by Cause

Analysis Methodology & Results

Research Question 1: How effectively can weather variables and seasonal patterns be used to predict wildfire occurrence?

Problem Statement:

Many weather metrics can potentially predict wildfires, but this section evaluates how effective they are when used with machine learning models.

Methodology:

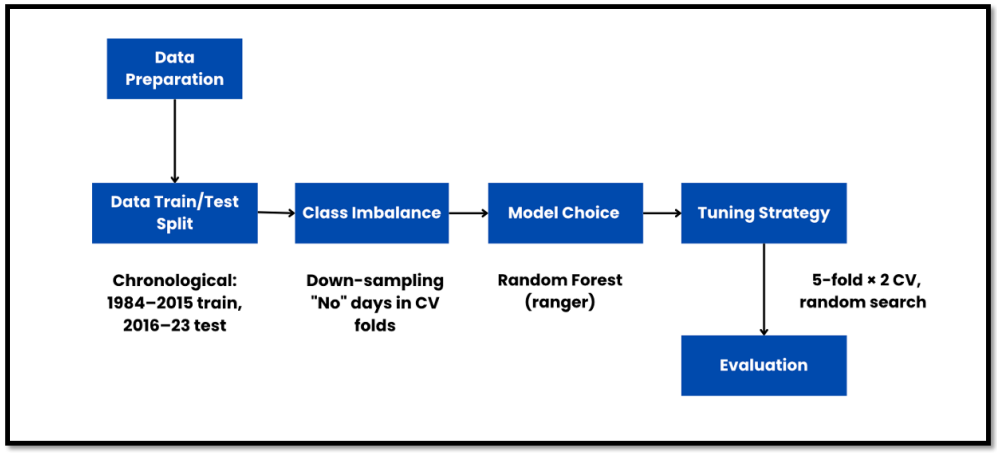

Figure 5: Flowchart

```
## [1] "Train dimensions: 11688 x 17"

## [1] "Test dimensions: 2922 x 17"

## [1] "Target variable distribution in training data:"

##
##   No  Yes
## 8061 3627

## [1] "Features selected for modeling:"

##  [1] "MAX_TEMP"              "MIN_TEMP"              "PRECIPITATION"
##  [4] "AVG_WIND_SPEED"        "TEMP_RANGE"            "WIND_TEMP_RATIO"
##  [7] "LAGGED_PRECIPITATION"  "LAGGED_AVG_WIND_SPEED" "SEASON"
## [10] "MONTH"

## [1] "Best Ranger hyperparameters found by CV:"
##   mtry  splitrule min.node.size
## 2    2 extratrees             1
## [1] "Ranger CV Results (for best tune):"
##   mtry min.node.size  splitrule       ROC      Sens     Spec       ROCSD
## 2    2             1 extratrees 0.8268372 0.7427118 0.787979 0.004318623
##       SensSD     SpecSD
## 2 0.01898514 0.02229162

## [1] "Test AUC: 0.8419 | Optimal Threshold: 0.4416"
## [1] "Confusion Matrix (Test Set):"
##           Reference
## Prediction   No  Yes
##        No  1220  316
##        Yes  358 1028
```

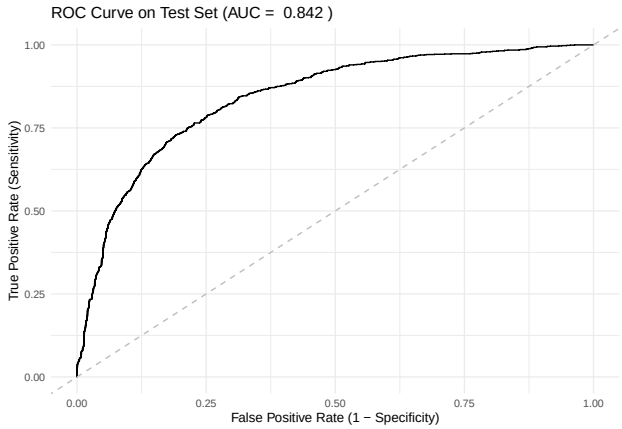

Figure 6: Figure : ROC Curve on Test Set.

Higher sensitivity helps identify true fire days.

This RF model generalizes well on unseen data (2016-2023).

Key Drivers:

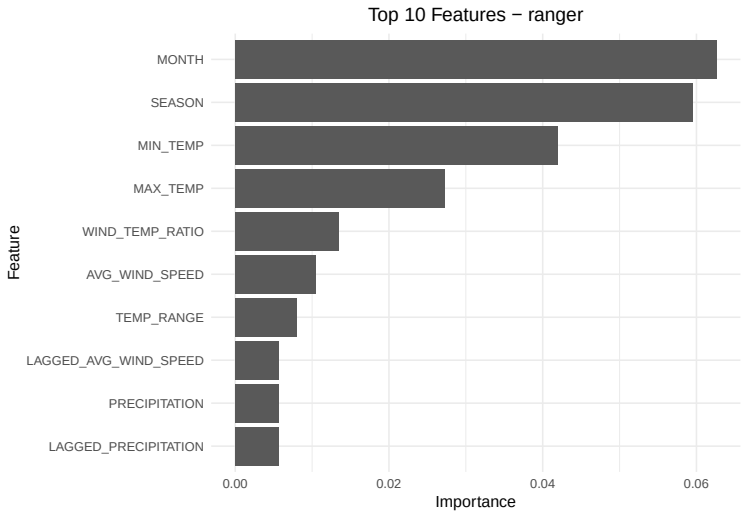

Figure 7: Top 10 Feature Importance

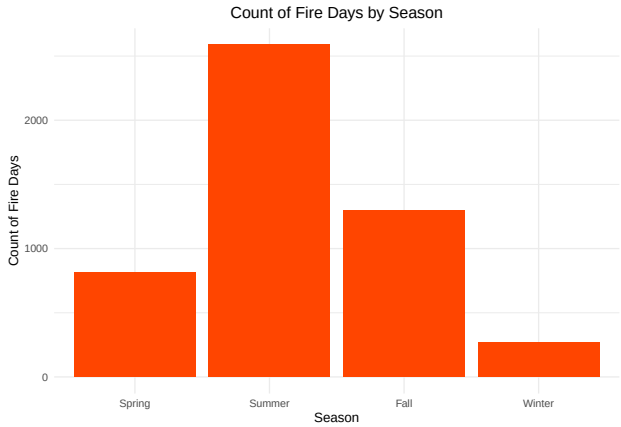

Figure 8: Count of Fire Days by Season

Conclusion:

This shows the efficacy of using ML models on the weather dataset to train and analyze various factors. This also indicates that weather and calender features can effectively predict wildfire ignition.

Month and temperature are better indicators of fire in comparison to other factors.

Research Question 2: Can We Categorize Unknown Causes of Fires into Known Causes?

Problem Statement
About 31% of fire data cannot be classified due to loss of evidence. We aim to classify unknown fires using ML models.

Method 1: Multiclass Classification Random Forest and XGBoost used. Agreement in classifications marked as true. Else, high-confidence predictions are accepted.

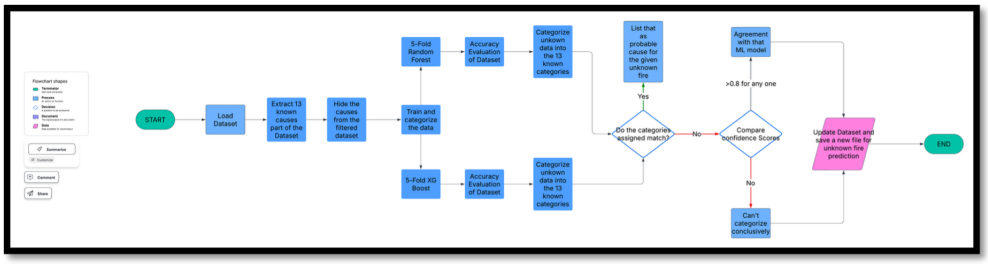

Figure 9: Flowchart

Results
~51.9% data classified.

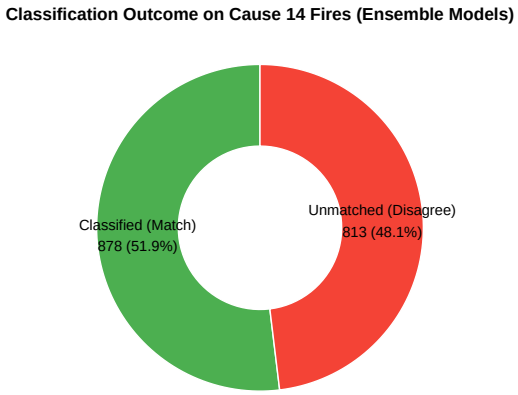

Figure 10: Method 1 Pie Chart

Metrics	5-Fold Random Forest	5-Fold XGBoost
Model Accuracy	39.39%	41.24%
Variable Importance	Maxwindspeed_kmh, Maxtemperatures_C, WindSpeed_kmh, Shape___Area, Temperature_C	Windspeed_kmh, FIRE_DURATION, Temperature_C, Shape___Area, Maxwindspeed_kmh

Conclusion
Low model accuracy due to class imbalance. Grouping fire causes into binary classes may improve performance.

Method 2: Kernelized SVM (RBF) Human-made vs. Natural fire causes.

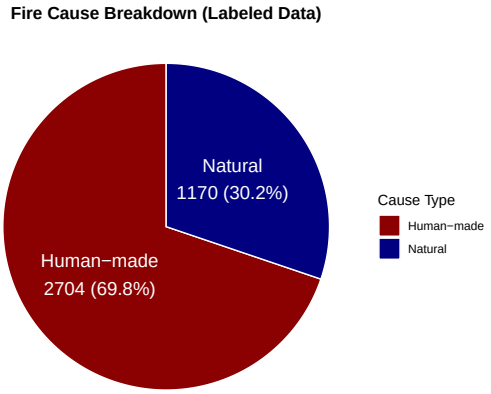

Figure 11: Natural vs Human Cause Split

Results
RBF kernel SVM tested with varying class weights.

Figure 12: Weight tuning performance

Final Metrics
- Accuracy: 84.99%
- Sensitivity: 61.95%
- Specificity: 94.20%

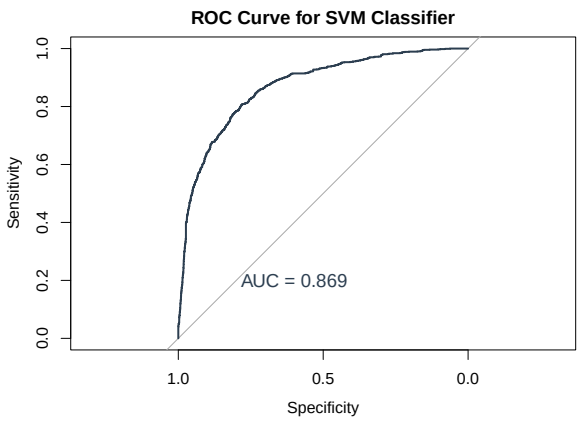

Figure 13: SVM ROC Curve

Conclusion
~80% confident classification; seasonal trend of predicted causes also follows pattern.

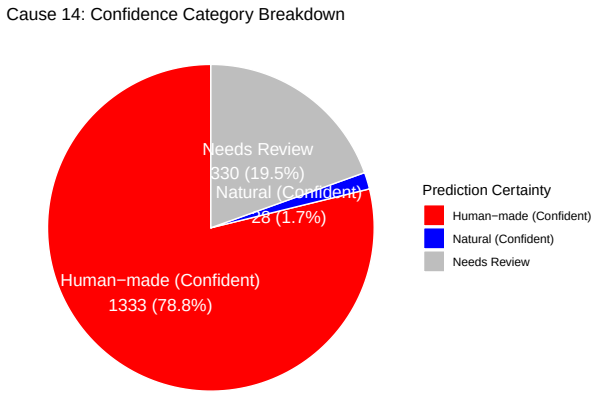

Figure 14: SVM classification plot

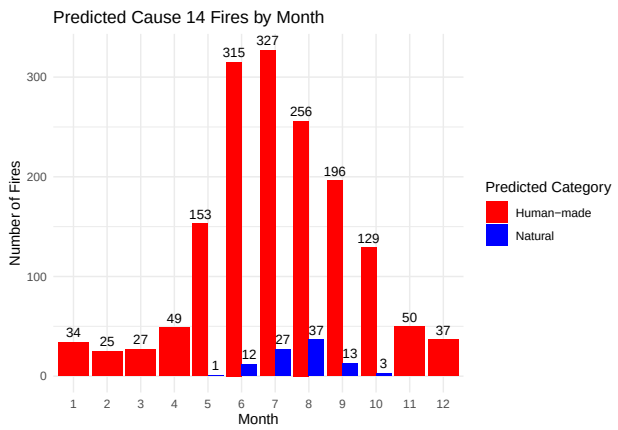

Figure 15: Seasonal Trend Reclassified

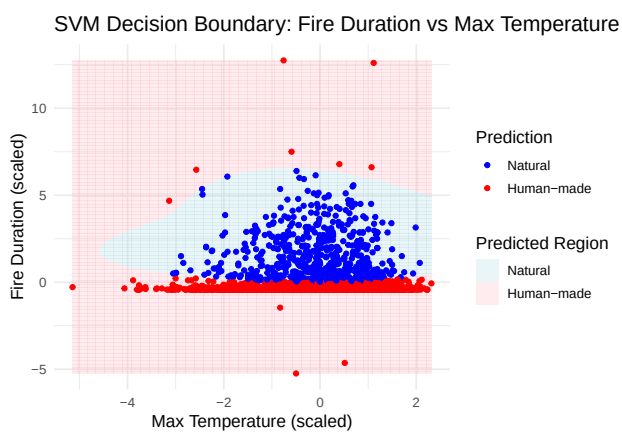

Figure 16: Example SVM grouping

Research Question 3: How Does Daily Weather Affect Wildfire Duration?

Problem Statement:

How do daily weather variables influence wildfire duration? We classify fire into two categories - short and long- based on most optimally accurate result.

Methodology: A 5-fold SVD with RBF kernel was used to train and test classification across fire durations, with 5 days identified as the optimal threshold for short vs long fires.

Results: Testing for optimal number of days:

Threshold sensitivity analysis (2-20 days) shows 5, 6, 7 days as good balance points. This is indicated by the higher kappa values achieved on these three days.

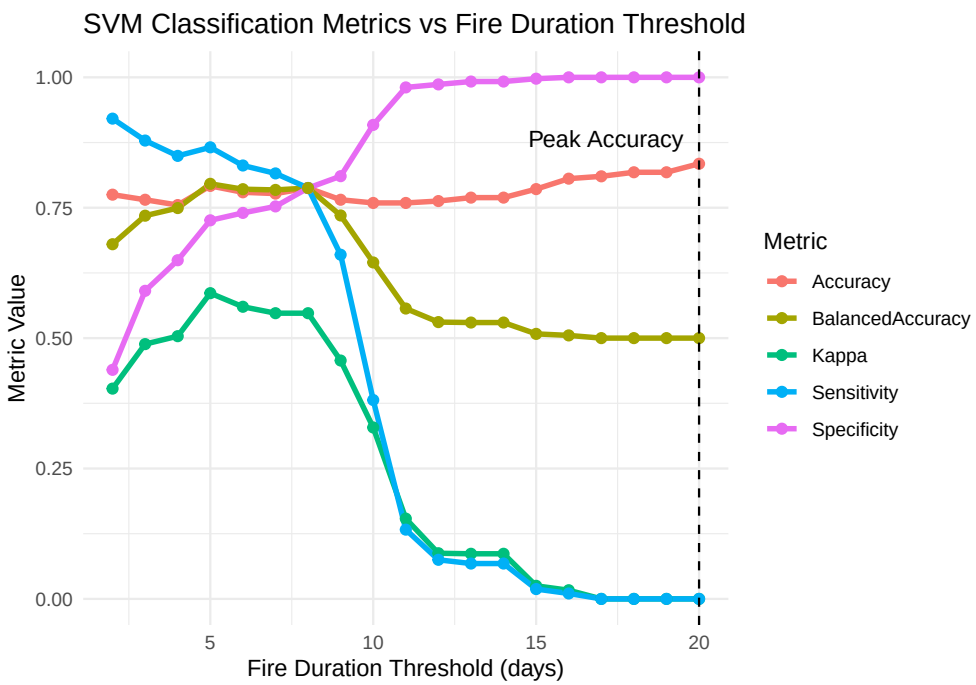

Figure 17: SVM Classification metrics vs fire duration threshold

Below figure compares the confusion matrices to help determine final class.

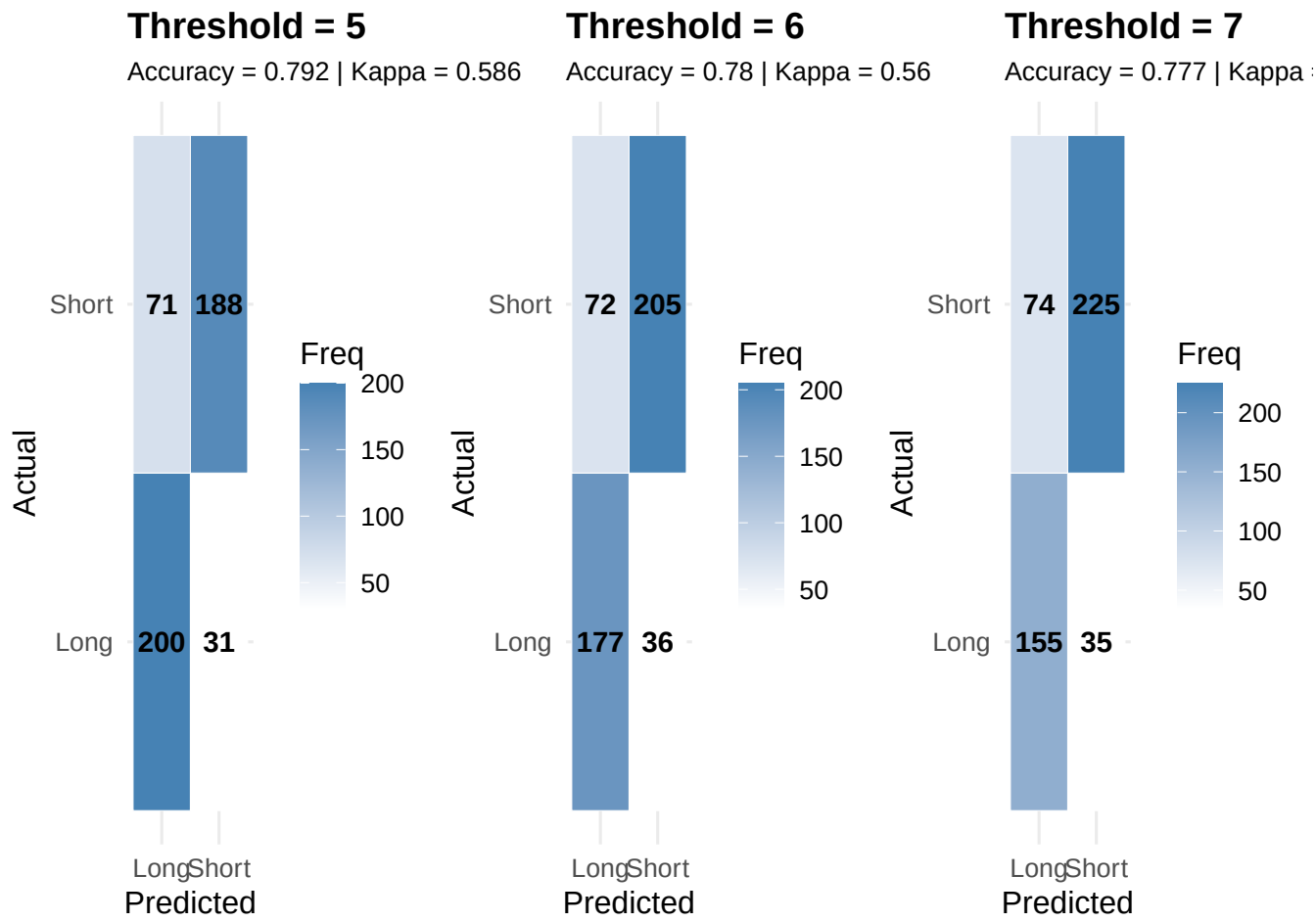

Figure 18: Confusion Matrices for Thresholds 5, 6, and 7

Metrics for n = 5 days.

```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction Long Short
##      Long   200    71
##      Short   31   188
##
##                Accuracy : 0.7918
##                  95% CI : (0.7532, 0.827)
##     No Information Rate : 0.5286
##     P-Value [Acc > NIR] : < 2.2e-16
##
##                   Kappa : 0.5862
##
##  Mcnemar's Test P-Value : 0.0001127
##
##             Sensitivity : 0.8658
```

```
##             Specificity : 0.7259
##          Pos Pred Value : 0.7380
##          Neg Pred Value : 0.8584
##              Prevalence : 0.4714
##          Detection Rate : 0.4082
##    Detection Prevalence : 0.5531
##       Balanced Accuracy : 0.7958
##
##        'Positive' Class : Long
##
```

- Overall Accuracy: 79.18%
- Kappa value: 0.5862

Conclusions:

SVM plots for boundary separation:

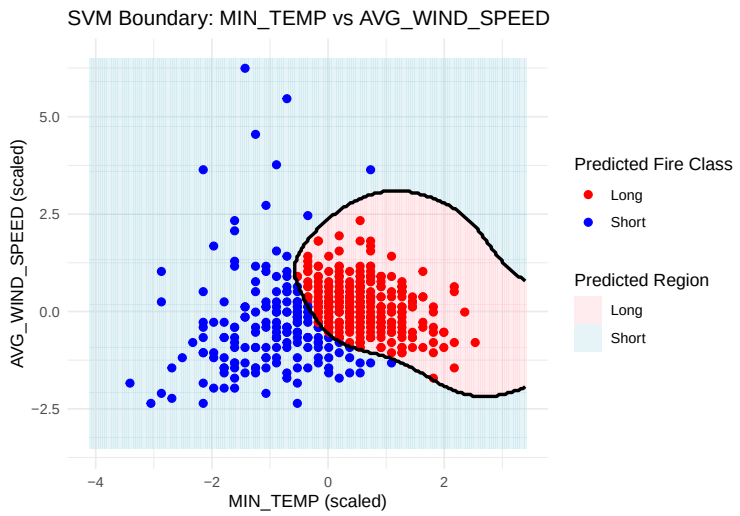

Figure 19: SVM Boundary: MIN TEMP vs AVG WIND SPEED

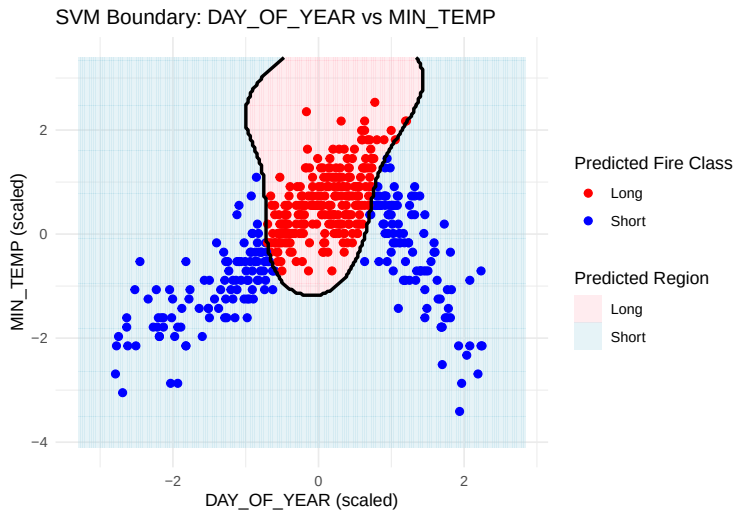

Figure 20: SVM Boundary: DAY OF YEAR vs MIN TEMP

- From fig19- longer fires usually have higher minimum temperature.
- From fig20- Longer fires are usually observed during middle of the year (May till July) and is usually distributed across higher maximum temperature.

  There are many such conclusions which can be derived from the trained model.

Linear models miss key weather patterns, while RBF-SVM effectively captures correlations for long-burn events, enabling resource prediction based solely on weather and temporal data.

Research Question 4: Wildfire Hotspot & Risk Zone Identification

Problem Statement:
Wildfires tend to cluster due to environmental factors; this analysis uses DBSCAN to identify geographic hotspots and risk zones for targeted prevention.

Methodology:
DBSCAN was applied to fire location data to identify dense wildfire regions. A composite risk score using temperature, duration, and wind speed categorized events into risk levels, visualized on a California map.

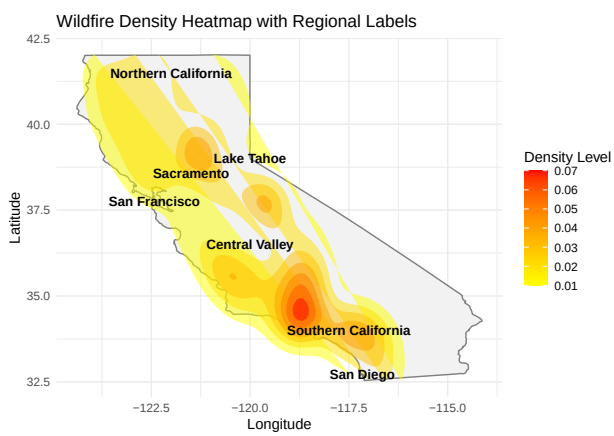

Figure 21: Wildfire Density Heatmap with Regional Labels

Southern California, Central Valley, and Sierra Nevada show higher fire clustering. DBSCAN was refined post-presentation for more robust verification.

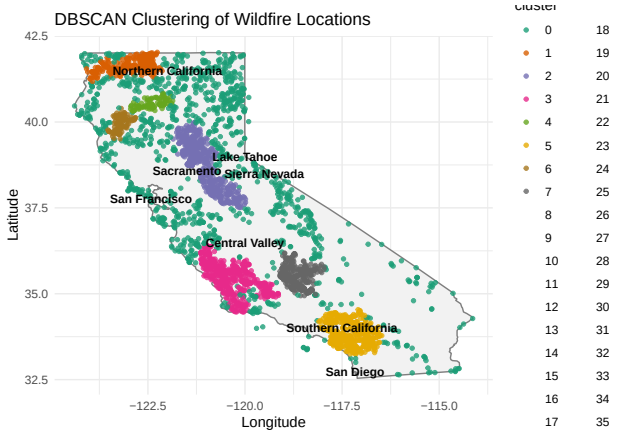

Figure 22: DBSCAN Clustering of Wildfire Locations

Next to evaluate fire behavior k-means clustering with k=4 was performed.

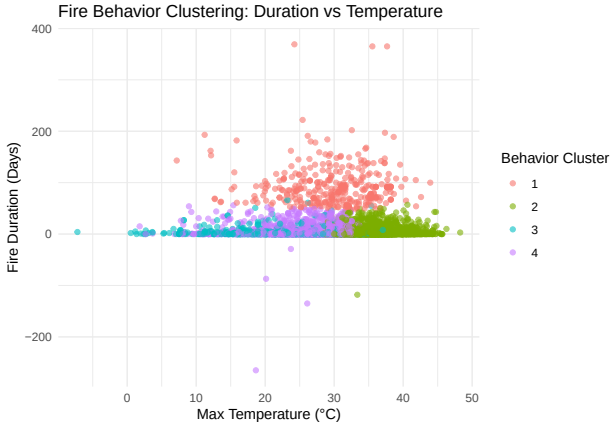

Figure 23: Fire Behavior Clustering: Duration vs Temperature

Above figure suggests, cluster-1 has higher fire duration and it leans towards average max temperatures of 30 deg-cel, which correlates with results seen in RQ3.

Further, weather statistics were evaluated for these 4 clusters.

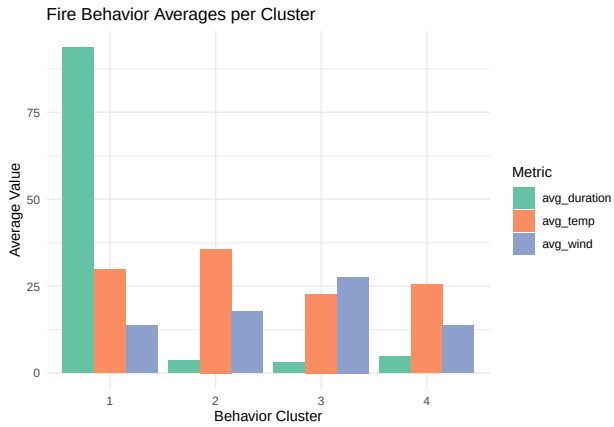

Figure 24: Fire Behavior Averages per Cluster

From above chart it can be seen that higher fire durations usually have higher temperatures but lower windspeeds, which shows windspeed may not have any significant impact on fire duration.

Fire Risk Map:

One of the most important outcomes from this research question is identifying the risk zones which are indicated in the map below.

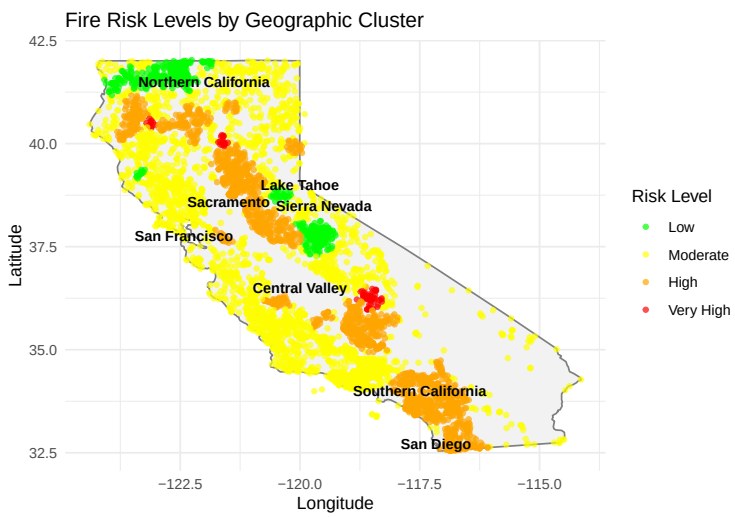

Figure 25: Fire Risk Levels by Geographic Cluster (DBSCAN)

High risk zones: Southern California, Central Valley, Sacramento.

Improvements

As suggested by instructor reviews and peer feedbacks following improvements were made after final presentation:

- Clear and cohesive arrangement for the flow of research questions.

- Results of using HDBSCAN for RQ4 is shown below:

  HDBSCAN results in a better clustering but required a lot of tuning and muting noisy values. Below figures indicate the same.

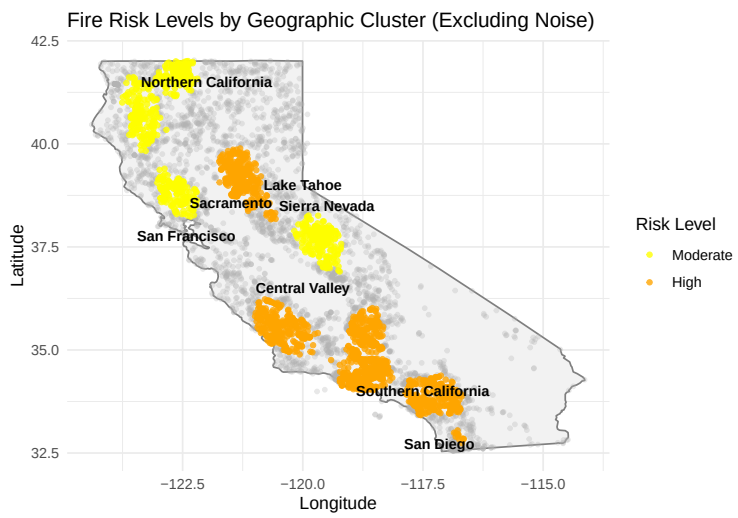

Figure 26: Fire Risk Levels by Geographic Cluster (HDBSCAN)

Above figure clearly shows, the risk zone mapping using HDBSCAN provides cleaner clusters.

Final High risk zones: Southern California (LA, Ventura) and Central Valley (Sacramento).

Future Work

- Exploring creation of web based dashboard for prediction of probable fire regions based on all the models trained above, by integrating live weather data to predict the zones of highest probable fire.
- Integrate projected monetary loss calculation based on the average property rates by area.

Contribution Statement

All team member contributed equally in exploratory analysis, data engineering, analysis and reporting. Weekly meetings were conducted to review progress.

References & Acknowledgements

1. CALFIRE Fire Perimeters
2. Zenodo Dataset
3. https://www.ncei.noaa.gov/access/monitoring/monthly-report/fire
4. https://oehha.ca.gov/sites/default/files/media/downloads/climate-change/document/04wildfires.pdf

We would like to thank Prof. Wang and Prof. Yu for teaching and guiding us through various concepts of statistical Data Science. We are very thankful for the timely feedback on the project which improved the quality of final implementation. We also extend our thanks to our classmates who supported us by providing critical and important feedback.

Addendum

Below are a few ChatGPT prompts that were used to optimize codes and generate results:

1. Why is the correlation between variables so low?
2. Why is this error occurring? Can we train the data from scratch instead of using filtered data?
3. How do I use this model to classify new data without re-training it every time?
4. Give me code to overlay decision boundaries and actual predictions in one plot.
5. What is the best way to tune class weights to optimize sensitivity?
6. Give me a standardized script to evaluate both manual and automated weight tuning using the same test split or folds.
7. How do I plot the ROC AUC curve for SVM?
8. How do I use this model to map all cause-14 based on the ML model we have?
9. I need to write a .Rmd file for the final report. How am I supposed to do that?